Exp No. 11: : Implementation of IoT-based Smart Agriculture using MATLAB

AIM: To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

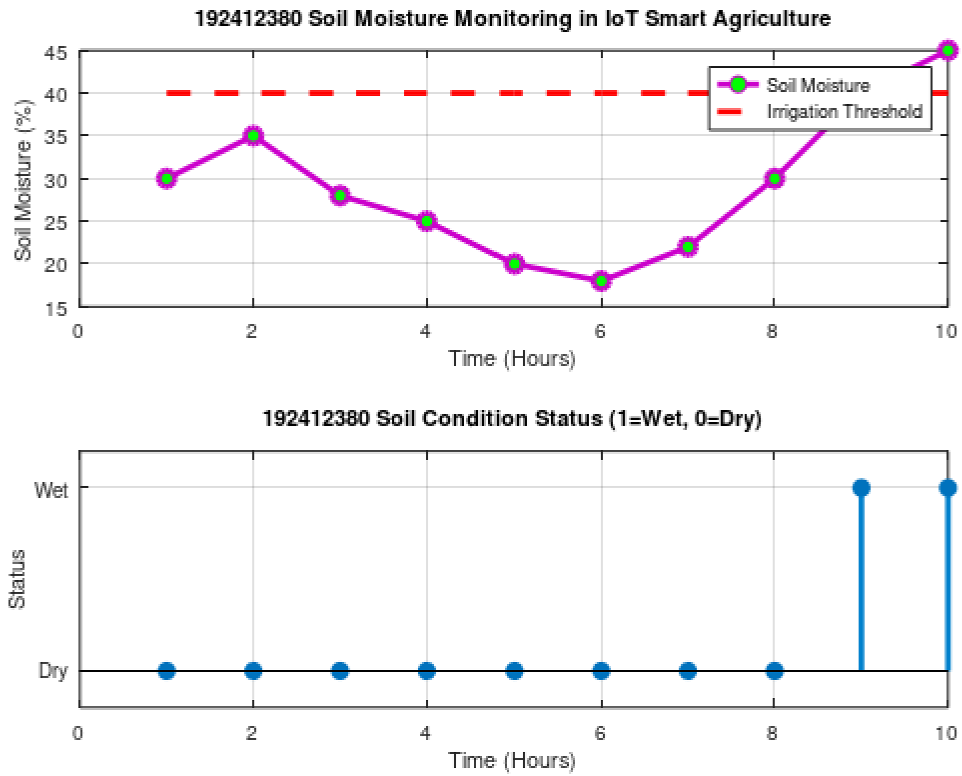
Objective 1:

To monitor the **Soil Moisture (%)** level in agricultural fields using IoT-based sensing simulation in MATLAB.

Program :

```
clc;
clear;
close all;

% Time in hours
time = 1:10;

% Simulated soil moisture (%)
soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold for irrigation
threshold = 40;

figure;

% -------- Graph 1: Soil Moisture Variation --------
subplot(2,1,1);

plot(time, soil_moisture, '-o', ...
    'Color', [0.8 0 0.8], ...
    'MarkerFaceColor', [0 1 0], ...
    'LineWidth', 2);

grid on;
hold on;

% Irrigation threshold line
plot(time, threshold * ones(size(time)), 'r--', 'LineWidth', 2);

title('192412380 Soil Moisture Monitoring in IoT Smart Agriculture');
xlabel('Time (Hours)');
ylabel('Soil Moisture (%)');

legend('Soil Moisture', 'Irrigation Threshold');
```

OUTPUT :



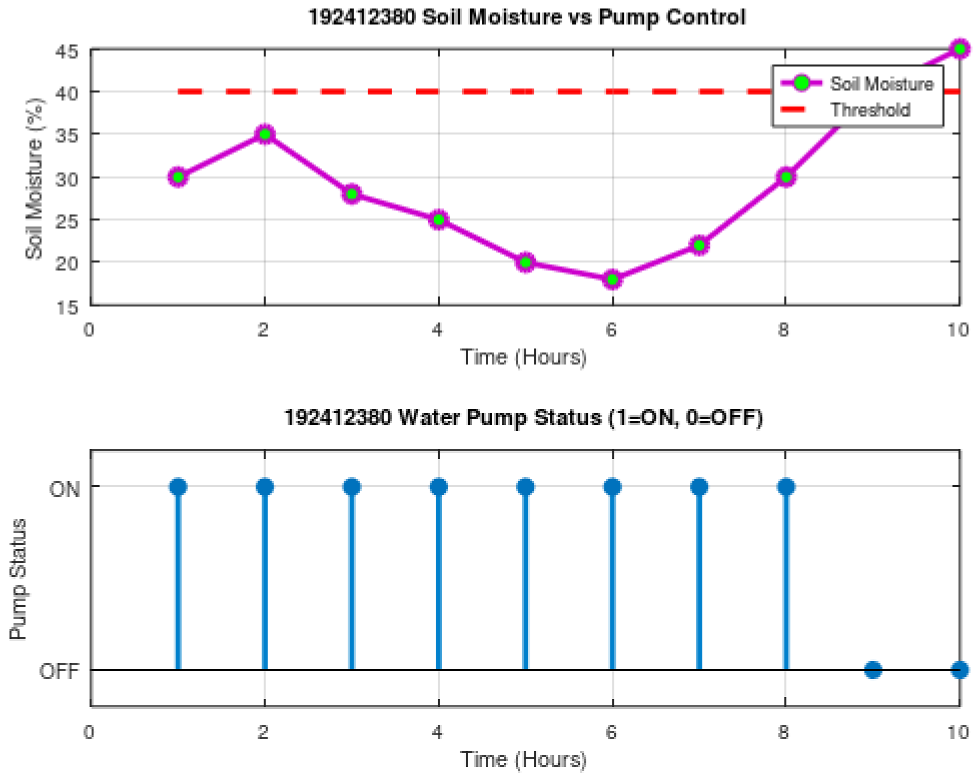
Objective 2:

To control and display the **Water Pump Status (ON/OFF)** based on threshold soil moisture conditions.

Program

```
clc;
clear;
close all;

% Time in hours
time = 1:10;

% Simulated soil moisture (%)
soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level
threshold = 40;

% Pump control logic
% 1 = ON, 0 = OFF
pump_status = soil_moisture < threshold;

figure;

% -------- Graph 1: Soil Moisture vs Pump Status --------
subplot(2,1,1);

plot(time, soil_moisture, '-o', ...
    'Color', [0.8 0 0.8], ...
    'MarkerFaceColor', [0 1 0], ...
    'LineWidth', 2);

hold on;

% Threshold line
plot(time, threshold * ones(size(time)), 'r--', ...
    'LineWidth', 2);

grid on;
```

OUTPUT :



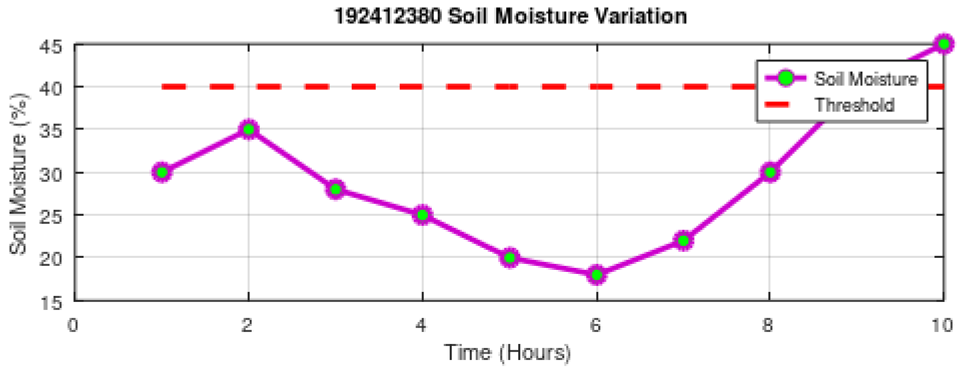
Objective 3:

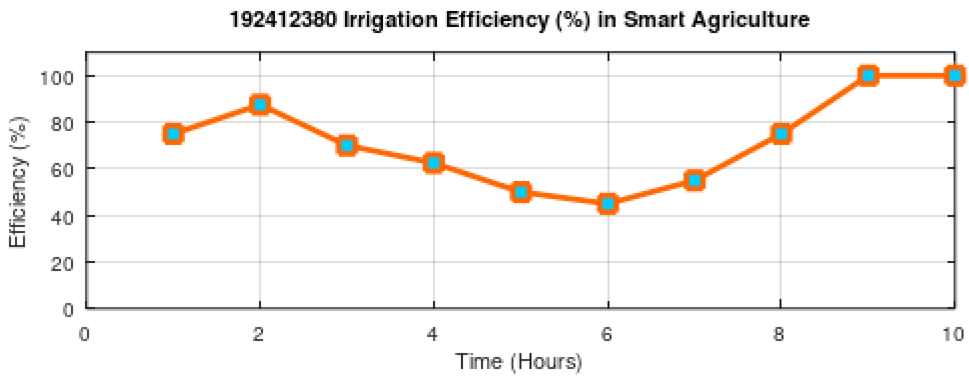
To evaluate the **Irrigation Efficiency (%)** by analyzing water usage effectiveness and moisture maintenance level.

program

```
clc;
clear;
close all;

% Time in hours
time = 1:10;

% Simulated soil moisture (%)
soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level
threshold = 40;

% Irrigation efficiency calculation (%)
efficiency = (soil_moisture ./ threshold) * 100;

% Limit efficiency to 100%
efficiency(efficiency > 100) = 100;

figure;

% -------- Graph 1: Soil Moisture vs Threshold --------
subplot(2,1,1);

plot(time, soil_moisture, '-o', ...
    'Color', [0.8 0 0.8], ...
    'MarkerFaceColor', [0 1 0], ...
    'LineWidth', 2);

hold on;

% Threshold line
plot(time, threshold * ones(size(time)), 'r--', ...
```

OUTPUT :





Result:

1. The soil moisture levels were successfully monitored using MATLAB simulation, showing variation over time and identifying dry and wet conditions based on threshold values.

2. The water pump control system was effectively implemented, where the pump automatically switched ON when soil moisture dropped below the threshold and remained OFF when moisture was sufficient.
3. The irrigation efficiency was successfully calculated and analyzed, showing that the system optimizes water usage by maintaining soil moisture close to the desired level.

Practical Questions

An IoT-based smart agriculture system is designed to monitor soil conditions and automate irrigation using sensor data. Using MATLAB, simulate the system to analyze soil moisture variation, control water pump operation, and evaluate irrigation efficiency.

1. To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.
2. To control and display the Water Pump Status (ON/OFF) based on soil moisture threshold conditions.
3. To evaluate the Irrigation Efficiency (%) for optimal water usage and improved agricultural productivity.